

Evidence grading protocol

Version 0.2.1 · the rulebook, rendered from evidence-protocol-v0.2.1.yaml

2026-08-17 · draft - for review by Ryan and a clinician reviewer (unappointed) · all evidence questions. Run so far: Q3 (exercise, diet, gut health), Q4 (medication - levodopa timing, wearing-off, disease modification, impulse control, tremor).

Change record

0.2.1

Three proposals accepted (q4-run-note, approved by Ryan 2026-08-17), all three surfaced by running Q4 rather than by review of Q3. P-8: W-8c split. It was firing on 55 of 110 Q4 studies, mostly because the paper could not be reached rather than because it carries no funding statement, so it was penalising studies for our retrieval. W-8c now applies only to a confirmed absence in a text we actually read; new W-8d flags an unreachable statement at zero points and lifts on retrieval, following W-11's precedent. P-9: new governance rule G-7 requiring claims to be stated as positive assertions. Six Q4 claims arrived phrased as negatives, where a Priority 3 grade asserted the opposite of the evidence. P-10: design_base declared for meta-analyses of observational studies at 3, matching the value the control panel had been using undocumented since v0.1. Q3 re-graded: no claim moves. Q4 re-graded: see the v0.2.1 diff shipped with this version.

0.2.0

Seven proposals accepted (protocol-v0.2-rule-proposals, reviewed and approved by Ryan 2026-08-15). New: claim lanes L-1 (P-1); external- evidence rules X-1/X-2 (P-2, P-3); non-extractable-results modifier W-11 (P-4); stacking rule S-1 (P-5); active-comparator blinding W-3c and claim-level body-of-evidence B-1 (P-7). Workbook change: D17 MEDI-PD re-graded Priority 2 -> 3 on its null primary (P-6). Engine-vs-workbook agreement 30/36 -> 37/37 (fibre row now gradable via X-2). Full rationale and per-row predictions in protocol-v0.2-rule-proposals.pdf, which is this version's G-3 log.

0.1.1

Tier labels renamed from advice language to priority language (per a community reviewer's comment, 2026-08-13: 'act on it' reads as medical advice). Tier 1 'Act on it' -> 'Priority 1'; tier 2 'Promising' -> 'Priority 2'; tier 3 'Not supported' -> 'Priority 3'. Grading rules unchanged; no study changes tier. New governance rule G-6: standard disclaimer required on all patient-facing artifacts. Per G-3, this diff ships with the next answer.

0.1.0

Initial draft, 2026-08-11.

SR - source registry

What is allowed to count as evidence, and at what trust level.

S_authoritative	members: cochrane_reviews, clinicaltrials_gov, anzctr, who_ictrp role: Registries are the single source of truth for trial status and pre-declared outcomes. Cochrane outranks other syntheses.
A_peer_reviewed	members: journals_indexed_in_pubmed_or_embase role: Standard evidence. Journal-level check against predatory list (Beall successor lists) before admission.
B_preprint	members: medrxiv, biorxiv role: Admissible for horizon-scanning. Grade capped (see C-3). Flagged 'preprint - not peer reviewed' wherever shown.
C_grey	members: conference_abstracts, theses, registry_only_results role: The excavation layer's raw material. Enters as leads; graded only if it survives extraction against the schema.
D_community	members: member_reports, patient_forums role: Hypothesis generation ONLY. Never graded as clinical evidence. Feeds excavation (E-7) and the future member loop.
X_excluded	members: press_releases, news_coverage, vendor_content, supplement_marketing role: Never evidence. Usable only as pointers to a primary source. (The '70% flavonoid survival' figure died here in Q3.)

L - claim lanes

Every claim is assigned a lane at extraction. The lane decides which grading scale applies.

L-1_intervention	question: does X help? rule: Grades under design_base + modifiers + caps unchanged.
L-1_prevalence_safety	question: how common is X? is X hazardous? rule: Observational designs are the appropriate designs here, so C-1 does not apply. SR/meta-analysis of adequate observational studies bases at 5; single observational studies at 3. W-8c records its flag against confidence but does not subtract (consistent with its own v0.1 note). All other modifiers apply.
L-1_risk_association	question: is X linked to getting PD? rule: Grades under existing rules, always displayed with the 'about risk, not progression' wording.

DS - design base scores

Starting score by study design, before modifiers.

meta_analysis_of_rcts	5
meta_analysis_observational	3
rct	4
prospective_cohort	3
case_control	2
cross_sectional	1
mechanistic_or_animal	1

W - weights and modifiers

Applied to the base score. Each has an id so disagreement can be specific.

W-1_replication	value: 1 when: independently replicated by a second group
W-2a_small_n	value: -1 when: n < 50
W-2b_large_n	value: 1 when: n > 500 or pooled n > 1000
W-3a_double_blind	value: 1 when: double-blind with credible placebo/sham
W-3b_open_label	value: -1 when: open-label or no blinding
W-3c_active_comparator	new_in: 0.2 rule: In trials where participant blinding is impossible by the nature of the intervention (exercise, diet, physiotherapy): W-3b does not apply if outcome assessors were blinded and the primary outcome is objective or rater-scored; W-3b applies in full if assessors were unblinded or the primary outcome is self-reported. W-3a and its bonus are never available to such trials. Replaces the control panel's unofficial 'strict blinding' toggle.
W-4_secondary_only	value: -1 when: claim rests on a secondary outcome also: cap at tier 2
W-5_surrogate	value: 0 when: outcome is a surrogate (imaging, biomarker) also: cap at tier 2
W-6_heterogeneity	value: -1 when: meta-analysis with I2 > 50%
W-7_below_mcid	value: -1 when: effect smaller than minimal clinically important difference
W-8a_interested_funder	value: -2 when: funder sells or markets the intervention AND design is weak (open-label, surrogate, or secondary outcome). Strong design neutralises: a preregistered double-blind industry RCT keeps its score, disclosed.
W-8b_funder_in_conduct	value: -1 when: any funder (including nonprofits) participated in design, conduct, analysis, or manuscript - as disclosed. Triggered by Barichella 2016.
W-8c_no_disclosure	value: -1 when: the full text was obtained and carries no funding or COI statement note: counts against confidence, not against the finding; common in pre-2010 papers. Revised in 0.2.1 (P-8): absence may only be recorded from a document we actually read. Where the text could not be obtained, W-8d applies instead.
W-8d_disclosure_not_retrievable	value: 0 new_in: 0.2.1 when: a funding or COI statement could not be obtained because the source itself was unreachable (paywall, captcha, robots block, dead link) also: flag 'funding statement not retrieved'. No points. Resolves to W-8c or to a recorded funder the moment the text is obtained. Like W-11, this is a statement about our access, not about the study, and it must never cost a study points. [new in 0.2.1]
W-9a_single_centre	value: -1 when: single centre and unreplicated

W-9b_geo_cluster	value: 0 when: positive results cluster in one country/region while multicentre trials elsewhere are negative also: flag prominently (live example: FMT - positive single-centre Asian trials, negative multicentre European trials)
W-10_attrition	value: -1 when: dropout > 20% without adequate handling
W-11_not_extractable	value: -1 new_in: 0.2 when: key result cannot be fully extracted and verified from accessible sources (paywalled effect sizes, inaccessible supplements, figures without data) also: cap at Priority 2; flag 'result not fully verifiable'. Cap and point lift the moment the numbers are obtained and verified - this is a statement about our verification, not the study.

S - stacking rules

Where two modifiers are symptoms of the same underlying limitation.

- S-1 When W-3b and W-9a both fire on the same study, apply -1 in total, not -2, and show both flags. They are symptoms of the same underlying thing: a modest single-site study without the resources for blinding or multicentre design. Each still applies at full value alone. W-2a remains independent - sample size is a distinct weakness.

X - external evidence

When general-population evidence may enter a Parkinson's question.

X-1_external_replication	rule: For a symptomatic, non-PD-specific outcome (the symptom and its treatment mechanism are not altered by PD in a way that would change the direction of effect), Cochrane-level general-population evidence for the same intervention and outcome counts as replication: W-1 applies, W-9a is neutralised, and C-4 is waived - provided at least one adequately designed PD trial confirms the direction of effect. Always displayed with the flag 'graded with external evidence', naming the external source. The PD-specificity judgement is recorded at extraction and is itself reviewable. Trigger case: macrogol (G4).
X-2_external_only	rule: A claim resting entirely on external general-population evidence (no confirming PD trial) is gradable when the X-1 non-PD-specificity condition holds, but is capped at Priority 2 however strong the external evidence, and carries the flag 'no adequately powered PD trial - graded on external evidence'. If the condition does not hold, the claim stays Context. Trigger case: fibre (G8).

C - caps

A cap cannot be bought back by bonuses.

C-1	max_tier: 2 for: any observational design, regardless of size
C-2	max_tier: 2 for: pilot / feasibility trials
C-3	max_tier: 2 for: preprints
C-4	max_tier: 2 for: single unreplicated RCT, unless multicentre + preregistered + significant on primary outcome

T - tier mapping

priority_1	label: Priority 1 gloss: strongest evidence min_score: 5 plus: no cap in force; risk acceptable; effect noticeable
priority_2	label: Priority 2 gloss: promising, unproven min_score: 3
priority_3	label: Priority 3 gloss: not supported by current evidence rule: score < 3, OR pooled RCT evidence is negative for the claim. A negative meta-analysis outranks any positive single trial (this is what settles FMT and boxing).
pending	rule: definitive trial registered and unreported - never anticipate results
context	rule: background science and guidelines; not graded

B - claim-level grading

- B-1 Where a claim is supported by two or more independent RCTs and a concordant meta-analysis, the claim takes the grade of the pooled body of evidence, and W-2a is assessed against pooled n rather than against each trial alone. Individual rows keep their own scores; the claim-level grade is what the answer cites. Writes down what 'the grade attaches to the claim, not the study' already implies. Trigger case: progressive resistance training (E5, PRET-PD).

F - funding and conflict rules

- F-1 Extraction of funder, funder role, and declared COI is a mandatory schema field for every study. Revised in 0.2.1 (P-8): the status is recorded as one of three values, never as a blank and never as a bare 'not found'. found - a statement was obtained and is recorded. absent - the full text was obtained and carries no statement (W-8c). not_retrievable - the source could not be reached, so nothing can be said either way (W-8d). A study may only be marked 'absent' on the basis of a document that was actually read. Where extraction was blocked, the honest value is 'not_retrievable', and it is expected to change when retrieval improves.
- F-2 Funding is always displayed beside the finding. It never silently excludes a study - transparency, not censorship.
- F-3 Penalties only via W-8a/b/c. The trigger is the combination of interest and weak design, not industry funding per se.
- F-4 Any tier-1 claim in a published answer must carry its funding disclosure into the synthesis, not just the table.
- F-5 Non-financial bias is assessed alongside: researcher allegiance (a modality's inventor trialling it), geographic clusters (W-9b).

E - excavation layer

Where non-obvious findings live.

- E-1 Registry sweep: trials completed > 2 years ago with no linked publication. Unpublished nulls change pooled answers.
- E-2 Registry results database: outcome data posted to ClinicalTrials.gov that never reached a journal.
- E-3 Preprint monitor for the question's terms (medRxiv).
- E-4 Conference abstracts (MDS Congress and similar) - results surface here 1-2 years before journals.
- E-5 Contradiction mining: where community practice diverges from evidence (boxing), or evidence diverges by geography (FMT). Each divergence is a research question nobody has framed.
- E-6 Cross-study subgroup mining: signals consistent across studies but headline in none (sex differences in the diet cohorts).
- E-7 Community-reported interventions with no literature at all - the member loop as hypothesis generator. The one source competitors cannot copy.
- rule Excavation output enters as leads (source class B/C/D), passes the same extraction and grading as everything else. No shortcuts to the answer.

G - governance

What makes this controlled rather than vibes.

- G-1 Semantic versioning. Any rule change bumps the version.
- G-2 Every published answer is stamped with the protocol version that produced it.
- G-3 Rule changes are logged with rationale, and all affected claims are re-graded; the diff ships with the next answer.
- G-4 Clinician sign-off is a release gate for anything patient-facing. Reviewer: unappointed (open item).
- G-5 No ad-hoc overrides. A grade that feels wrong means a rule is wrong: propose the rule change, apply it everywhere, re-run.
- G-7 Claims are stated as positive assertions, phrased so they could be true or false, with the Priority reading as the strength of evidence FOR the claim as written. 'Drug X slows progression' graded Priority 3 says the evidence does not support it. 'No drug slows progression' graded Priority 3 says the opposite of what its own evidence shows. Introduced in 0.2.1 (P-9) after six Q4 claims arrived phrased as negatives and inverted their own meaning. Where a claim is restated to satisfy this rule, the restatement is recorded on the claim, and the change is a claim edit rather than a rule change.
- G-6 Every patient-facing artifact carries the standard disclaimer, verbatim: 'Priority ratings rank the strength of published evidence for each claim. They are not medical advice and are not instructions. Decisions about medication, diet or exercise belong with you and your treating team.' Wording changes to the disclaimer are rule changes (G-1/G-3). Final wording subject to clinician review (G-4).

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